

ActInf Textbook Group ~ Cohort 6 ~ Session 13

(Chapter 6, Part 1) ~ 7/15/2024

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all right welcome back to cohort Andrew take away really quickly can you all hear me okay perfect um yeah so hey everyone welcome back uh cohort 6 just resuming here Midway through July uh we're going to be starting the second half of the textbook uh first half we were introduced to a lot of the underlying ideas and mechanics behind active inference uh Bas role Basi and brain hypothesis uh hidden States being inferred from observations uh action as a means of trying to change the hidden State through transitions uh a lot of really kind of foundational ideas uh towards understanding active inference as well as following along uh the low road active inference starting from basian can statistics as well as the hod active inference following the uh principle of free energy minimization so the second half of the textbook I think is um it's great for better understanding how to more directly apply active inference so chapter six a recipe for Designing active inference models right off with that kind of gives you a sense of what the rest of the textbook is actually going of potentially be about we'll be learning how to design models uh chapter seven and eight will be about uh putting together discret and continuous time models respectively um so that's great for people who want to um you know figure out which kind of model they need for their particular use case it's nice that there are rather expansive chapters on both types of those models chapter 9 uh gets more into analysis so once you've set up a an active inference simulation uh it starts to discuss like how you can actually interpret your results um depending on your use case like how you might set things up depending on what your particular question or inquiry is whenever you're setting up the simulation and finally chapter 10 is a nice uh wrap that is maybe slightly out of place compared to the rest of what I said about uh the second half of the book but it's uh it's good it's it's a rather holistic kind of uh conclusion type chapter both wrapping up the book comparing active inference to a variety of other disciplines and Fields like control theory uh other social sciences psychology uh reinforcement learning as another approach these kinds of problems and so on so uh just a quick overview on chapter six again a

recipe for Designing active inference models uh the book the uh the intro to this chapter uh friston at all WR uh the challenge is not to emulate the brain piece by piece but to find the generative model that describes the problem the brain's trying to solve the model provides it so that refer two is the model provides a complete description of a system of interest and the resulting Behavior inference and neurodynamics can all be derived from a model by minimizing free energy all these ideas should be familiar from the first half of the book but it's just more clearly stating yes active inference is heavily inherited from or derived from uh different ways of approaching computational Neuroscience that said we are are not trying to recreate you know um billions of individual neurons and all of the kind of crazy uh sparsely connected architecture underlying say the human brain or otherwise the emphasis upon developing generative models um you know this kind of gets into the difference between complexity and accuracy found in uh any machine learning related problem reinforcement learning problem so on so again not trying to recreate the entire brain uh just to where we can get things accurate enough to where may start to track and potentially could even be fitted to empirical data um so whenever uh we'll be introduced over time to more Exemplar kind of models but we can adapt the form of the models presented in this textbook to our respective problem of Interest we can change their form uh should they be shallow or hierarchical we can modify the variables involved I.E uh the beliefs so what are the states to infer what are the observations that the agent receives what are the actions or policies available to the agent to commit so from there there's it's really just a four-step recipe um there are entire discussions that could be had about each step but in summary uh first part first step is decide which system are we modeling so what are the internal versus external States what are the sensory versus action States uh what is the marov blanket of the system what separates it out are there nested marov blankets involved here those are all basically questions and then step two of the recipe is deciding what's the most appropriate form for the generative model like I mentioned earlier chapter seven and chapter eight which discuss uh discret versus continuous time models so that's a big decision to make uh potentially involving somewhat different computations for minimizing free energy underneath and one of them were're uh working with categorical probability distributions another one we're working more with uh Taylor series and approximations uh all of which are covered in the textbook as well um we know that hierarchical models versus shallow models so hierarchical models May operate at a slower time scale uh they might uh basically involve learning states are beliefs about states which are then supplied back down to a lower layer uh supplying prior for the lower layer at the next

time step or otherwise um other questions involve like the depth and of inference and of planning how far into the future is the agent uh inferring States s questions of like expected free energy come up so uh step three of the recipe how do we set up the generative model well we've already touched on it but we decide on the model's variables States observations actions um in more detail that means what are the observation modalities what are the H State factors overall um other things like how do we Define the agent's habits or the prior over its available policies um that's kind of its starting point for deciding how to act uh what if we exclude all the factors from the equation and then finally which parts of the model are fixed and which parts are learned so certain parts might be updated over time While others remain uh static uh variables in the model and finally step four how do we set up the generative process uh the quick and easy way of saying that is how do we Define everything that is not the agent right so so if it's the agent who commits actions uh well those actions are then going to technically be received by the environment is one kind of figurative way of looking at it so you need to Define an environment that can take in actions and then at each step whether in discrete or continuous time output the appropriate observation um which will then be received by the agent so that starts to define a active inference action perception Loop whatever setting things up in code so um that's in general the the sum of the chapter as far as its four steps go there's plenty of other things that are discussed along the way and those four steps broken down into uh smaller parts but um I think that's a good enough overview for now so I'll open up the floor for any kinds of questions or just general points of interest that people might want to bring up to start a conversation about and then I believe Daniel had some some other thoughts regarding other projects that are going on Institute uh depending on what we have time for I really enjoyed the uh the first point the which system are we modeling because as we design a system that's going to have some specific environments we're going to have some problem that we may have imagined solving inside of it which of course we're going to set up our Markov blanket and we're going to set up our uh all of our situations but it was interesting to me because it seems like that's exactly what nature did whenever it built like the brain and us and stuff like that and it kind of put it into perspective that the resolution in which you're trying to solve is kind of an arbitrary decision of variables and where you arbitrarily decide to set up your Markov blanket and it kind of just set it into motion to me that oh it's just Markov blankets all the way down yeah definitely I always anytime I I start to reflect more on Markov blankets and what's been written about them I just feel like we're in a world of uh of Relativity um you know if I attempted to model like myself say like What Separates Me from what's

around me um well that said I'm composed of a series of smaller markov blankets right I'm composed a series of neurons and and and other sort of processes that potentially could be seen as as engaging in active inference themselves or other inferential processes so um yeah it's it's so important to be able to delimit if you're going to actually apply active for instance set up a simulation like what separates what is your system of interest and what separates it from those so-called um otherwise we don't we don't know what to do we don't know what to code and um yeah we don't quite know what our field of studies yeah going off of that um I was just reading some of RX environments documentation earlier today um and in in that case they actually treat the environment or you can treat the environment um like it's its own agent in essence right it just doesn't have a generative model behind it but you know that's abstracted away um just going off the Markov blankets all the way down yeah a lot of lot of great points just on that last RX environments piece it seems like part of what is happening here is the generalization of entity and then subtypes for like agent and Niche and and that highlights what but is also um like the Deep symmetry between agent and Niche with the same blanket one person's send and receive as the other person's receive and send in like a um Alice and Bob communication situation um and then also I mean there there's there's a lot of other um pieces to say like if you do account for both directions across the interface like you have modeled what is at the overlap of the agent in the niche so that point choosing like where which part belongs to which it's like it's it's almost defined as the interface belonging to both or interfacing with both so that leads to a lot of like the more philosophical speculations and and discussions around markov blanket while while also just pragmatically giving a very nuanced way to to talk about like nested systems absolutely I found it really I'm sorry please go ahead I was just gonna say practically just it eases implementation a lot in the case of Arts I found that really interesting and I also wanted to ask guys I can't seem to find it in the text embarrassingly right now but whenever it was talking about creating an agent that could walk they were saying that you could have an agent that embodies some generative model that learns how to walk or you can have legs that are built with the Dynamics that in inherently don't have a generative model but know how to walk already know how to move in that environment I found that to be really interesting because it asked for me the question of do we need the generative model or do we just need a generative model that has the generative model and is that just a larger generative model in of itself do you guys think about that at all do you have any thoughts one one relevant Direction brought up earlier was that like the choices about the model can always be cast as a higher order modeling topic and then another piece

which I'll find the link for will involve like this kind of kind of embodied modeling like um let's just say that the the elbow has like a limited amount of lateral Mobility or something like that then if you're going to make a map of that control theory setting there might be two kinds of maps you can make one that just simplifies no there's only one um control variable and it has like this limit and has this kind of like frictions and things like that so over Evolution that would come to embody features that would make the first order inference easier and easier but that's not a guarantee of just throwing some code on the page but perhaps in in iterated evolution with embodiment features of the model that in silica would be purely needing simulation would come to like off Source or just be heris sixs for what the embodiment was already doing yeah yeah I think it um comes back to I I personally view it as like that kind of complexity accuracy trade-off question again um one of my favorite uh stimulations from active inference like pieces of literature I've read I'm personally interested in I guess what would be called computational Psychiatry and uh there's this nice paper um that was put together to try and simulate a classic tze Paradigm but where the agent has like an effective layer um all that is to say is that you you take the t ma Paradigm you model a mouse that has a hierarchical layer that relates to emotion and see how the the mouse's uh Behavior changes in the teamas and and how it learns differently and I mean super cool and that said there are dozens upon dozens of like model architecture decisions that could have been made there uh you've added an entire aspect to the model and you could have you know there could be so many different things that you included in this so why why just that one thing suddenly is what the the question that comes next rather than thinking know this is a massive addition um so yeah I it's just a lot of it is relativity I think um classic way of going about things is to design a model um do your best and if you're designing it for a particular use case uh see if it's possible to try and fit that model to empirical data through trying to you know approximate behavior of a real um living organism or if you're trying to produce something that you know has very high optimal behavior for a particular task Robotics and so on so just have to test things out this is that kind of one of the examples it's like interesting how it's it is being pursued in duramorph and other areas like this is this is a great empirical example through differential stiffness and and control on differential stiffness you have multi- joints action with a falling weight with differential stiffness and like different tune tuning stiffnesses but that's how posture actually is held rather than being like this wide open control space like well my my shoulder could be at any angle and my elbow could be any angle with then then there's going to be many control like dimensions and interactions um so another another um topic to

connect here um is an example so the so in the uh summary Andrew gave of step three how to set up the generative model like what we associate to what this is kind of a link to the o inference ontology this is why we do learn a generic system independent scale-free ontology even though our pragmatic interest might be like in this specific you know X um here's taking basically what are coming up in chapter 7 the discrete time generative model as the a c d e o s by listing those and then here it's at three nested levels of modeling like the pherone level most um feet on the ground detecting phone perception that was kind of what we got into in the infra ants paper but then also we could think okay well but the location switching and that's kind of like the live stream 42 slam where the bottom level of the nested model was the posture and the top was location so you could have like a warehouse robot that was like I want to go like to location on the other side but not have to do all the postural modeling just having a slower transition and then here it's like okay well and task could have a transition maybe it's just a binary transition or that would be the exploration how you model it but this is the whole point about the system and scale independent ontology is that it's not the details of the system which is why the ontology doesn't have opinionation on like how many nestmates should be in a colony or how how rapidly should nurses shift the forgers or back and forth like th those are once you kind of see it very far in the last mile of modeling that system of interest and so it's totally described by chapter six why are we making this model what are we actually choosing to model what what kind of features are we taking as data what Latent states are we going to consider but the tabular form helps like structure what otherwise can be a very open-ended assembly process I guess this is all it just clarifies it visually and this is also good input for language models with templates of other code that's uh that's really fascinating and it also kind of reminds me of a problem that I heard of in self-driving cars from a y a researcher named yon Le and it was in his paper about uh jeppa which they're asking how you create a self-driving car that pays attention to important things like other cars on the road instead of just focusing on in machine learning just classically minimizing the loss because if you give really really high definition cameras to the environment the car is going to start looking where the most information content is which is going to be at the trees because those leaves moving there is so much information going into there but you're completely focusing on the wrong things if you're looking at trees whenever you're driving you might just crash so what they have to do in choosing the right sort of variables is they found out they have to use these hierarchical models which to try to cast it into active inference terms they had to create a hierarchy of a few different Markov blankets however the actual variables for the interfaces of

the inward blankets had to be kind of coalescing inwards they had to be abstractions you aren't allowed to use the same variables again and again because then you would find yourself in a recursive problem where you keep looking at the trees instead of focusing on things like time scales or important variables that occur down the road and that's just something that I thought was tangentially related to uh setting up the generative model really interesting stuff yeah great comments a few ways that's played out so here's like one one um fan favorite paper from 2021 that makes a three-level phenomenological model so this is commonly replicated and you're right like a nested model does instantiate a Markov blanket such that let's just say that experience were residing in the green but it doesn't even matter about whether it's phenomenological like awareness without going too deep into that from the green perspective the the the node that interfaces with the orange is the marup blanket and then also it's like that actually goes further and so it is the reason why although sometimes the discussion is like what is the Markov blanket quote around in agents but at the same time s13 is blanketed from s23 by b b is the markof blanket intervening between two time steps it's the transition operator or a the this the observation to Hidden State mapping like the partially observable part is the Markov blanket between the observable and the hidden state so like the markovian time concept past only influencing the future through the present that's just the mark I'll blanket through time so it's it's not a statement or expression about H how any agent or any Baye Network should be it's actually so pervasive that the content itself does like relatively limited work because it's just a feature of any base graph that's not fully connected is going to have these markof blankets so it it doesn't like distinguish certain kinds of sparse connected Bas graphs from others it it's just a feature that is like basically is the sparsity of the graph that is absolutely fascinating I can't wait to check that out thank you for bringing it up yeah that that's a cool one and then another another angle to to like bring a more recent paper um than than the textbook is uh active data selection and information seeking which was live stream 57 with with friston and Par in the dot one and then par joining in for do two um and and great preparation work for thezero like that is one of the things that's super exciting about active inference is there can be a lot of useful tinkering along the low road and like generating different architectures for for nest and models and unified approaches to to perception cognition action Etc here are some explicit ways to Value the information with epistemic value and that gets into talking about pragmatic plus epistemic value but this is the difference between reinforcement learning and active inference in reinforcement learning epistemic features must get projected down to or onto or get lumped in with utility reward because

there's like a uni ranking that different policies are getting ranked against and then policies are selected or sampled from that utility basis and so there's all kinds of tricks like well give a bonus for learning information but that would also come to exactly the same question which is how should information be valued and then like a sort of first pass approach like just have high you know frame rate differencing which is the origins of predictive coding where the frame rate differencing is giving High information content that's where you would want to focus but that is like a rapidly moving tree like you brought up like it could have high frame rate differencing but it wouldn't reduce uncertainty about is there a tree there and so active data selection and considering epistemic value explicitly helps give exact and composable epistemic values for generative models and gives like a playground slash space where epistemic and pragmatic value are not the same as each other and yet they can be balanced so there is the parametric balancing like if it's an explore exploit Tas or multi-arm Bandit it's not just like making the situation is going to guarantee that you have resolved the tradeoff even though language is kind of used to that extent or effect in the textbook that this like dissolves the explore exploit trade-off whereas a slightly more toned down approach that does doesn't like promise a given result is it sets up the setting where explore exploit can be compared without collapsing like the explore component to some higher order exploitation problem just a quick comment on that that um I've personally been uh putting together a three-hour tutorial on applying active inference I'll actually be uh presenting it in a couple days uh International Conference on computational social sciences and Philadelphia a couple days from now and I I've you know one of the strongest points of interest for social scientists is this uh explore exploit you know dilemma and I just I think one of the most interesting points about is it can almost be viewed as like both things are being included in roughly the same optimization problem and I I think that the textbook is coming out of a context where where this active inference is being heavily compared to reinforcement learning is a potential alternative way of going about producing these models um and exploration and exploitation do tend to be treated separately where you need like an additional set of rules uh for defining like what what is it that defines exploratory behavior for an agent versus exploitative and it seen I think is just introducing additional assumptions uh into a reinforcement learning model in order to differentiate those whereas with active inference there's this sort of quote unquote natural balancing between the two but also agreed that like yes now in the same optimization problem they're both included in U Computing expected free energy for an active inference agent but it doesn't necessarily mean the agent is always going to find some theoretical correct

amount of exploratory value or Tim value in particular action versus pragmatic there could be
 n there could be zero it could always think like oh if I do this or this I'm never going to learn
 anything so I might as well just go for the goal so to speak um but yeah sorry I've just been
 thinking about that a lot while uh putting this together but maybe that's useful for for someone
 thinking about it yeah where that's shown most uh clearly in analytically figure 26 here's
 expected free energy it's a functional on Π which is like a vector of policy priors that sums to
 one and then actions are either drawn from the top or the most likely or sampled from Π with
 either a straight sampling on Π or you can have a temperature on action so sample from a
 more blurred out P or a more sharpened Π what G does in these models expected for energy is
 it takes a policy prior prior to the update and then updates it into a policy posterior so it's
 sharpening policies according to pragmatic plus epistemic value both explicitly designed so it
 has these elements 1 2 3 4 five but really it's two major terms the first one here is pragmatic
 value so if you only have pragmatic value only utility maximization that is expected utility and
 basian decision Theory so this does not Encompass there to be learning or at a higher level it
 reflects a setting where a modeler has given a utilitarian value to a learning event so in
 practice that might pursue learning like learning likeed Behavior but it it did so by saying well
 uh getting a problem right is worth one but learning how to do it is worth two so then pursue
 as many points as you can and then it starts learning capacities or something or you can have 2
 three four five included and drop the price magmatic value this is like the novelty search
 Infomax because it it kind of has implicitly a flat utility distribution so then decisions are
 weighted only according to the amount of information that they provide which is
 operationalized as the K Divergence double line between hidden States X Tilda with π either
 way and Y tlda is the sequence of observations so if the sequence of observations was
 expected to contribute nothing to beliefs about extra time this K Divergence would be zero
 whereas if yelda is anticipated to change beliefs on X through time that has a high KL
 Divergence so it's the Pure epistemic Element the pure pragmatic element is about alignment
 of observ future observations through time with preferences the prior Distribution on
 observations and then those two pragmatic and epistemic imperatives come together in
 expected free energy and then there's the degree of freedom of how those are balanced and
 then choices about how would you dynamically allocate the balance so it's not solved in
 practice however analytically it's pretty clear well another fun direction to to at least um begin
 here is like we're two years after publication of the book and one of the biggest changes in the
 active inference environment has been RX and fur um several of us are are working on this

and [Music] um we have uh figure 1.2 with with help from some of the bias lab I I'll put this in the chat and and we'll see but the the goal of this kind of um shared project between the ARX and forur project and textbook group the goal is to have code for every setting and figure in the textbook and that would be um we can do it in a few different repos we can have uh copies in uh notebooks so it can just be run uh there's a table of all RX and fur examples so here's the reactive base these are 23 examples from a documentation here's kobus eser Housen here's eight examples from his is website learnable Loop then here's figure 1.2 so here's a notebook that will install you click run runtime run all first time it hits the first block it's going to install Julia then it'll say reload the page that brings you into the Julia kernel then runtime run all again second time through it installs it runs the tests that confirm that Julia is installed and then it installs the packages that are relevant for RX and fur then here is is a great example of without even going into RX and verse performance engineering or like computational benefits just focusing on the textbook group possibilities and and educational possibilities here we have figure 2.1 so it's like the Frog of the apple is either jumping out of the hand or not and then we have an agent who has priors about the prevalence of frogs and apples in the world and then they update what they believe they're actually holding on to or did did hold on to based upon the observation either a jump or not so the at model Block in RX and fur is like the not to crack this is the closest not not the closest it could ever be but this is the closest we've ever seen the concordance between the equations that analytically Define a generative model and the code to do so and so that also captures the biggest difference with RX and fur from a design perspective overall which is in PDP and in in mat laab there's a procedural definition of for example the agents interfacing with the environment like for a new observation then this happens that kicks off this calculate this update that move this to here and so it gives a script like flow that's a manable to explicit looping over like elements of planning or elements of of across multiple agents then there's a lot of degrees of freedom with like okay but then you know should agent one go first every time step or how how do we deal with that ARX infer replaces that with a declarative programming style so here this is the base graph as declared and it's basically what you see is what you get with the types of distributions and they could be parameterized here directly or it could be parameterized elsewhere and then the inference is also as direct as you can get it's basically just calling that base graph as declared doing some stuff under the hood that we can explore more and then letting that graph in its definition be uncoupled from the actual message passing like Logistics of information flow between the nodes that's the part that goes under the hood now with RX and fur so it opens up this

possibility where the app model specifications of the generative model look a lot like the equations and like the minimal way to describe a situation without this extra control flow of which nodes are updated in what order however that has to be supplemented in so that's a lot of the ongoing work is like what are the syntax and possibilities for these app model functions what else needs to come into play to to have interpretability and and kind of like getting it run without any errors also but getting it working in a in a way that seems to function and maybe even Leverage The reactive message passing part and then also what that would mean is not just for like different languages and programming backgrounds and everything but then in this par textbook group there would be a really clear path from like here's figure or 2.1 and now here's how you could continue to learn more about RX and fur and that would take one like all the way to the Horizon of modeling it in advanced settings so that's a clear pathway whereas the 2022 textbook otherwise the appendix doesn't fully reproduce all the figures it's also in mat lab so then it's like well you're out of for with whether you recommend or go to study these examples code in more detail or kind of start on a fresh late with like PDP which is probably what in earlier cohorts we discussed a lot but now there's kind of a tiebreaker with RX and fur and the missing artifact that we could greatly benefit from would be the one-time finite basically task of making RX and for models and notebooks for the textbook group a and in doing so making the accessory um kind of Explorations and having the conversations that are like the questions in chapter six and so just kind of then then we like we have like a tinkering format in common and one that we can run in notebooks and one that looks largely like the equations kind of written out more in uh programming form so not not not to go too too much on this but and and this notebook also contains some extraneous um information but if we can Bridge it with RX and fur and the textbook group It'll like really give a great connection between the education onramps and the application that that is also like an open source framework yeah I would I would second that um I think that'd be super cool um I've personally been using uh PDP for that tutorial I I mentioned earlier um and I mean the developers of it are are great um you know I've been talking with him personally a little bit um but that said yeah I've seen a lot of the benefits of RX andur and hope to get back into the RX andur group uh seems like really exciting work and finally to bring it back um I think especially for part two of the book at for anyone anyone in the textbook groups coming in cohorts looking to actually design active inference models themselves like this chapter like what how do we Define a system of Interest how do we actually code that is the the additional question that isn't necessarily answered uh in the chapters that we're reading here right it's kind

of implied in that you know it's uh the authors of the textbook do use mat lab as you mentioned Daniel and so it's kind of as well as the the the appendix does include like some uh some figures of direct matlab code constructing the a matrices and so on and PP but yeah being able to supplement the textbook group with RX in for code which is also from my perspective the most up to-date way of maybe even doing active inference I think I think that would be incredibly exciting and I think it would also Supply people with one a great Learning Resource like oh this is how you do this and then two uh IT Supplies a bunch of potential templates that people could download and modify for their own personal experiences um in the sense that you know part of active inference is that we don't just learn by changing our perception we learn by changing the world so um making these kind of templates available for people for them to try and test out their ideas themselves I think kind of fulfills the other half of the equation of maybe how to learn how to apply place so really exciting stuff yeah I just wanted to Second the value of templates for ARX and fur in that different models can be hierarchically composed um along with each other so yeah once one is specified has a lot going forward yeah getting some really clear documented examples and and getting the visualization functionalities and then the examples of what those motifs look like building libraries that can then be referenced by llms for code generation to um building some more methods for like the meta programming and and the interpretability moving in and out of other formats like there there's a lot of exciting ways to go um and chapter six is going to be kind of there meeting point because that's where we will that's that's the beginning of the modeling process again with the selection of the system of Interest which is kind of this like open ended stage but then following that it starts to progressively specify and eventually in a way constrain the output of just a final generative model so that that Dem mystifies I think a lot of the process and and will will like create lower friction paths to get from like a given data set to something that's that's doing something useful I remember one discussion from from many cohorts ago um of like then there's the pr Matic value of making the generative model which would be like okay I'm making air conditioning and so I prefer it to be like low energy so then I need to make a working model so that it works and then there's also the epistemic value like people learning about active inference learning about thems learning about each other um like learning about what is possible that it's like it never even needs to reach the final artifact although that final artifact might also provide epistemic value separate from its pragmatic value too so it's like not a worthless process e even if it um and and and how hard different stages are it's super system dependent and that Frontier is very rapidly moving too so it's it's

quite different than it was like one or two or three years ago so I would expect it to be also multiple fold to orders of magnitude easier in one two three years so it's just sort of like jumping in while you can and then your own experiences and Records will help calibrate and show where things actually are shifting and how easy or hard they are okay I do want to quickly add because I know we about out of time but uh much much earlier uh during introductions uh someone I think Zach had mentioned an interest in interpretability of models and I would love to see in RX for like I'm saying this partially because of just ignorance of of what people have tried so far but of course people are minimizing free energy but maybe a little bit of emphasis on like different of helper functions that can maybe pull out you know for a given time step for example what was the expected Information Gain um of you know choosing a particular action or policy for the agent like basically having functions that help you pull out like well here's the parts of this that are comprehensible right um for me I'm putting together this tutorial for social scientists it's like well if we can't extract it any interesting or useful insights from these models then why are we memorizing all of this relatively complex math why are we spending our our time doing that and so like for me it's like oh well you can you know make say 30 agents and now you can have them do things that lead to emerging behaviors but now you can actually peer into you know if if they're learning their likelihood models now you can look at their particular beliefs and how they change over time and consider those to be something like cognitive metrics that we can actually interpret like we can interpret them mathematically we understand how they work but we also can see how they impact Behavior Uh model performance and Longevity um and and so on so yeah it'd be super cool to to just make sure with all these nice uh technical advancements that are being made with with Julian ARS Ander to to also supplement that with with you know interpretability and and having that is a ready Resource as well yeah no absolutely oh sorry after you oh yeah I was I was just gonna say I I think today it's it's probably quite limited but arxon for debug mode does show like messages being passed it's most useful to my understanding recommended um for when you're unable to optimize the the graph right you're not converging on on a solution you're just running forever um usually you're running some sort of intractable problem I don't know how much that helps with interpretability I don't understand message passing super well um and I I just swed a a talk from Demitri bagv earlier today it's link to the AR for the engineering Discord Channel um he talks about how it took him years to understand message passing so there's that um the other thing that I would I don't think this functionality exists but I would possibly look forward to maybe understanding these

models ESP as they growing complexity um is sort of like loss of function research on them so like disabling or removing parts of the subgraph um because these are all all of the message passing happens locally it's just in between two nodes um so you can you know the example that's commonly used is like you could lose a sensor that's feeding into the model um and it would still continue to work with it's just optimizing without that um so yeah maybe that that would be kind of an interesting approach as well I'll try to give the shortest possible message passing that this this is a super interesting paper I hope we can like understand these different approximations but but leaving that for a second the 2017 paper with with Bert here's the key figure this is connecting figure 4.3 discrete time generative model this is a base graph nodes are the random variables edges are the causal relationships in the 40 Factor graph the nodes and edges are flipped so that the node are like combiners or functions of where different variables which which are on the edges come from there's as as they showed there's like a one to one mapping between a base graph and a foreign factor graph so they can contain the exact same information what you gain though with a factor graph is this like Logistics or scheduling order so it allows you to like turn the crank ahead one step and to say that's going to take these 22 messages in this exact order what those messages are there's a lot of different like payloads they can contain but what it lets the whole graph do is do one descent on a variational free energy landscape for example it could probably also do other things so in 2017 they were like we're already making these Bas graphs here's a logistically deterministic scheduler that'll let us crank even on a large base graph let us crank the whole thing ahead one time step at a time that that's message passing or variational message passing the difference with reactive and what Demitri I think brought in with the rocket is it uncouples the update frequencies from different parts of the graph so instead of having to turn the whole crank and run this deterministic schedule of every single thing you can have the reactive like just in time sending and receiving I believe that opens up some other things that need to be specified like how the logistics are but it seems like those are just conventions and actually the reactivity means that like the graph doesn't have a order or Tempo per se but it's like figuring out all those things are some of the most important things right and this kind of reactivity um I I've wondered like the connection between message passing here and message passing in the case of uh like multi or or distributed computing um it seems like with the reactivity you're getting possibly very close to that I don't know if it's something that's been discussed much that's super interesting I think it's probably almost totally unexplored with a multi-threading there's multi-threading and concurrent uh programming as like a multi-agent situation right I don't

know if how like if those kinds of moves have been brought into like possible multi-cpu or GPU RX and fur yeah yeah I don't I I don't know either it's it's very interesting the parallel there it seems yeah I'd love to learn more about it but I I also know not too much about distributive Computing so maybe something to explore cool well this is a fun uh sort of reentry into six like also we'll have the three-week Rhythm so let's in the third weeks Let's either work on a textbook example Andor annotating code from from CIS or the examples but let let let's go into RX and fur and um and curate the questions along way and make the notebooks Etc possibly we could record these sessions but we'll see if it's just as more like an application Workshop um I think this will be really fun though and help G give like kind of a textbook part two plus because after having done and then also then we leave these kind of like perhaps multiple Workshop templates for cohort 7 going through the second half half and then that would be like a jumping off point for okay then you finished that but you're already pretty familiar with artics and Fur by the end of the textbook group because the application oriented part of the book gives templates and examples so still many people SL all people will be learning code and Julia and RX and fur but then that's where we can also connect to using llms for code explanations and all this other stuff so thank you Andrew thank you John and Zach thank you guys so much Daniel I'm excited to read those papers and check out our I'll talk to you guys soon see you bye